Chapter 3
Continued Fractions

The continued fraction decomposition of real numbers grows naturally from the Euclidean algorithm, and continued fractions have been used in some form for thousands of years. One goal of this volume is to show how they relate to a natural action on a homogeneous space. To start there would be to willfully reverse their historical development: We start instead with their basic properties(38) from an elementary point of view in Sect. 3.1, then show how continued fractions are related to an explicit measure-preserving transformation in Sect. 3.2. In Chap. 9 we will see how the continued fraction map fits into the more general framework of actions on homogeneous spaces.

Let us mention one result proved in this chapter. We will show that for every irrational $x \in \mathbb{R}$ there is a sequence of 'best rational approximations' $\frac{p_n(x)}{q_n(x)} \in \mathbb{Q}$, defined by the continued fraction expansion of x. Moreover, for almost every x we have

$$\lim_{n\to\infty} \frac{1}{n} \log \left| x - \frac{p_n(x)}{q_n(x)} \right| \longrightarrow -\frac{\pi^2}{6\log 2},$$

which gives a precise description of the expected speed of approximation along this sequence.

3.1 Elementary Properties

A (simple) *continued fraction* is a formal expression of the form

$$a_0 + \cfrac{1}{a_1 + \cfrac{1}{a_2 + \cfrac{1}{a_3 + \cfrac{1}{a_4 + \cdots}}}} \tag{3.1}$$

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which we will also denote by

$$[a_0; a_1, a_2, a_3, \dots]$$

with $a_n \in \mathbb{N}$ for $n \geqslant 1$ and $a_0 \in \mathbb{N}_0$. Also write

$$[a_0; a_1, a_2, \dots, a_n]$$

for the finite fraction

$$a_0 + \cfrac{1}{a_1 + \cfrac{1}{a_2 + \dots + \cfrac{1}{a_{n-1} + \cfrac{1}{a_n}}}}.$$

Thus, for example

$$[a_0; a_1, a_2, \dots, a_n] = a_0 + \frac{1}{[a_1; a_2, \dots, a_n]}.$$

We will see later that the expression in (3.1)—when suitably interpreted—converges, and therefore defines a real number. The numbers a_n are the *partial quotients* of the continued fraction. The following simple lemma is crucial for many of the basic properties of the continued fraction expansion.

Lemma 3.1. *Fix a sequence $(a_n)_{n\geqslant 0}$ with $a_0 \in \mathbb{N}_0$ and $a_n \in \mathbb{N}$ for $n \geqslant 1$. Then the rational numbers*

$$\frac{p_n}{q_n} = [a_0; a_1, a_2, \dots, a_n] \tag{3.2}$$

for $n \geqslant 0$ with coprime numerator $p_n \geqslant 1$ and denominator $q_n \geqslant 1$ can be found recursively from the relation

$$\begin{pmatrix} p_n & p_{n-1} \\ q_n & q_{n-1} \end{pmatrix} = \begin{pmatrix} a_0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a_1 & 1 \\ 1 & 0 \end{pmatrix} \cdots \begin{pmatrix} a_n & 1 \\ 1 & 0 \end{pmatrix} \text{ for } n \geqslant 0. \tag{3.3}$$

In particular, we set $p_{-1} = 1, q_{-1} = 0, p_0 = a_0$, and $q_0 = 1$.

Proof. Notice first that the sequence $(a_n)_{n\geqslant 0}$ defines the sequences $(p_n)_{n\geqslant -1}$ and $(q_n)_{n\geqslant -1}$. The claim of the lemma is proved by induction on n. Assume that (3.3) holds for $0 \leqslant n \leqslant k-1$ and p_n, q_n as defined by (3.2) for any sequence $(a_0, a_1, \dots)$. This is clear for $n = 0$. Thus, on replacing the first k terms of the sequence $(a_n)_{n\geqslant 0}$ with the first k terms of the sequence $(a_n)_{n\geqslant 1}$, we have

$$\frac{x}{y} = [a_1; a_2, \dots, a_k]$$

as a fraction in lowest terms where x and y are defined by

$$\begin{pmatrix} x & x' \\ y & y' \end{pmatrix} = \begin{pmatrix} a_1 & 1 \\ 1 & 0 \end{pmatrix} \cdots \begin{pmatrix} a_k & 1 \\ 1 & 0 \end{pmatrix}.$$

Then

$$\begin{pmatrix} a_0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x & x' \\ y & y' \end{pmatrix} = \begin{pmatrix} p_k & p_{k-1} \\ q_k & q_{k-1} \end{pmatrix} = \begin{pmatrix} a_0 x + y & a_0 x' + y' \\ x & x' \end{pmatrix},$$

so

$$\frac{p_k}{q_k} = \frac{a_0 x + y}{x} = a_0 + \frac{y}{x} = a_0 + \frac{1}{[a_1; a_2, \dots, a_k]} = [a_0; a_1, \dots, a_k],$$

which shows that (3.2) holds for $n = k$ also. □

An immediate consequence of Lemma 3.1 is a pair of recursive formulas

$$p_{n+1} = a_{n+1} p_n + p_{n-1}$$

and

$$q_{n+1} = a_{n+1} q_n + q_{n-1} \tag{3.4}$$

for any $n \geqslant 1$, since

$$\begin{pmatrix} p_{n+1} & p_n \\ q_{n+1} & q_n \end{pmatrix} = \begin{pmatrix} p_n & p_{n-1} \\ q_n & q_{n-1} \end{pmatrix} \begin{pmatrix} a_{n+1} & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} a_{n+1} p_n + p_{n-1} & p_n \\ a_{n+1} q_n + q_{n-1} & q_n \end{pmatrix}.$$

It follows that

$$1 = q_0 \leqslant q_1 < q_2 < \cdots \tag{3.5}$$

since $a_n \geqslant 1$ for all $n \geqslant 1$; by induction

$$q_n \geqslant 2^{(n-2)/2} \tag{3.6}$$

and similarly

$$p_n \geqslant 2^{(n-2)/2} \tag{3.7}$$

for all $n \geqslant 1$. Taking determinants in (3.3) shows that

$$p_n q_{n-1} - p_{n-1} q_n = (-1)^{n+1} \tag{3.8}$$

and hence $\frac{p_1}{q_1} = a_0 + \frac{1}{q_0 q_1}$, $\frac{p_2}{q_2} = \frac{p_1}{q_1} - \frac{1}{q_1 q_2} = a_0 + \frac{1}{q_0 q_1} - \frac{1}{q_1 q_2}$ and

$$\begin{aligned} \frac{p_n}{q_n} &= \frac{p_{n-1}}{q_{n-1}} + (-1)^{n+1} \frac{1}{q_{n-1} q_n} \\ &= a_0 + \frac{1}{q_0 q_1} - \frac{1}{q_1 q_2} + \cdots + (-1)^{n+1} \frac{1}{q_{n-1} q_n} \end{aligned} \tag{3.9}$$

for all $n \geqslant 1$ by induction.

This shows that an infinite continued fraction is not just a formal object, it in fact converges to a real number. Namely,

$$\begin{aligned} u = [a_0; a_1, a_2, \dots] &= \lim_{n\to\infty} [a_0; a_1, \dots, a_n] \\ &= \lim_{n\to\infty} \frac{p_n}{q_n} = a_0 + \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{q_{n-1}q_n}, \end{aligned} \tag{3.10}$$

is always convergent (indeed, is absolutely convergent) by the inequality (3.6). Moreover, an immediate consequence of (3.10) and (3.5) is a sequence of inequalities describing how the continued fraction converges: if $a_n \in \mathbb{N}$ for $n \geqslant 1$ then

$$\frac{p_0}{q_0} < \frac{p_2}{q_2} < \cdots < \frac{p_{2n}}{q_{2n}} < \cdots < u < \cdots < \frac{p_{2m+1}}{q_{2m+1}} < \cdots < \frac{p_3}{q_3} < \frac{p_1}{q_1}. \tag{3.11}$$

We say that $[a_0; a_1, \dots]$ is the *continued fraction expansion* for u. The name suggests that the expansion is (almost) unique and that it always exists. We will see that in fact any irrational number u has a continued fraction expansion, and that it is unique (Lemmas 3.6 and 3.4).

The rational numbers $\frac{p_n}{q_n}$ are called the *convergents* of the continued fraction for u and they provide very rapid rational approximations to u. Indeed,

$$u - \frac{p_n}{q_n} = (-1)^n \left[\frac{1}{q_n q_{n+1}} - \frac{1}{q_{n+1}q_{n+2}} + \cdots\right] \tag{3.12}$$

so by (3.5) we have[(39)]

$$\left|u - \frac{p_n}{q_n}\right| < \frac{1}{q_n q_{n+1}}. \tag{3.13}$$

By (3.4) we deduce that

$$\left|u - \frac{p_n}{q_n}\right| < \frac{1}{a_{n+1}q_n^2} \leqslant \frac{1}{q_n^2}. \tag{3.14}$$

Recall from Sect. 1.5 that we write

$$\langle t \rangle = \min_{q\in\mathbb{Z}} |t - q|$$

for the distance from t to the nearest integer. The inequality (3.14) gives one explanation* for the comment made on p. 7: using the fact that any irrational has a continued fraction expansion, it follows that for any real number u, there is a sequence (q_n) with $q_n \to \infty$ such that $q_n\langle q_n u\rangle < 1$.

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* This can also be seen more directly as a consequence of the Dirichlet principle (see Exercise 3.1.3).

Lemma 3.2. *Let $a_n \in \mathbb{N}$ for all $n \geqslant 0$. Then the limit in* (3.10) *is irrational.*

PROOF. Suppose that $u = \frac{a}{b} \in \mathbb{Q}$. Then, by (3.14),

$$|q_n a - bp_n| < \frac{b}{a_{n+1} q_n} \leqslant \frac{b}{q_n}.$$

Since $q_n \to \infty$ by the inequality (3.6) and $q_n a - bp_n \in \mathbb{Z}$ we see that

$$q_n a - bp_n = 0$$

and hence $u = \frac{a}{b} = \frac{p_n}{q_n}$ for large enough n. However, by Lemma 3.1 p_n and q_n are coprime, so this contradicts the fact that $q_n \to \infty$ as $n \to \infty$. Thus u is irrational. □

The continued fraction convergents to a given irrational not only provide good rational approximants. In fact, they provide *optimal* rational approximants in the following sense (see Exercise 3.1.4).

Proposition 3.3. *Let $u = [a_0; a_1, \dots] \in \mathbb{R} \smallsetminus \mathbb{Q}$ as in* (3.10)*. For any $n > 1$ and p, q with $0 < q \leqslant q_n$, if $\frac{p}{q} \neq \frac{p_n}{q_n}$, then*

$$|p_n - q_n u| < |p - qu|.$$

In particular,

$$\left| \frac{p_n}{q_n} - u \right| < \left| \frac{p}{q} - u \right|.$$

PROOF. Note that $|p_n - q_n u| < |p - qu|$ and $0 < q \leqslant q_n$ together imply that

$$\frac{1}{q} \left| \frac{p_n}{q_n} - u \right| < \frac{1}{q_n} \left| \frac{p}{q} - u \right| \leqslant \frac{1}{q} \left| \frac{p}{q} - u \right|,$$

giving the second statement of the proposition. It is enough therefore to prove the first inequality. Recall from (3.13) that

$$\left| u - \frac{p_n}{q_n} \right| < \frac{1}{q_n q_{n+1}}$$

and

$$\left| u - \frac{p_{n+1}}{q_{n+1}} \right| < \frac{1}{q_{n+1} q_{n+2}}.$$

By the alternating behavior of the convergents in (3.11), each of the three bracketed expressions in the identity

$$\left(u - \frac{p_n}{q_n} \right) = \left(\frac{p_{n+1}}{q_{n+1}} - \frac{p_n}{q_n} \right) - \left(\frac{p_{n+1}}{q_{n+1}} - u \right)$$

is positive (if n is even) or negative (if n is odd). It follows that

$$\left|u - \frac{p_n}{q_n}\right| = \left|\frac{p_{n+1}}{q_{n+1}} - \frac{p_n}{q_n}\right| - \left|\frac{p_{n+1}}{q_{n+1}} - u\right|,$$

so

$$\left|u - \frac{p_n}{q_n}\right| > \frac{1}{q_n q_{n+1}} - \frac{1}{q_{n+1}q_{n+2}} = \frac{q_{n+2} - q_n}{q_n q_{n+1} q_{n+2}} = \frac{a_{n+2}}{q_n q_{n+2}}$$

by (3.4) and (3.14). It follows that

$$\frac{1}{q_{n+2}} < |p_n - q_n u| < \frac{1}{q_{n+1}} \tag{3.15}$$

for $n \geqslant 1$.

By the inequalities (3.15),

$$|q_n u - p_n| < \frac{1}{q_{n+1}} < |q_{n-1}u - p_{n-1}|$$

so we may assume that $q_{n-1} < q \leqslant q_n$ (if not, use downwards induction on n).

If $q = q_n$, then $|\frac{p_n}{q_n} - \frac{p}{q}| \geqslant \frac{1}{q_n}$, while

$$\left|\frac{p_n}{q_n} - u\right| < \frac{1}{q_n q_{n+1}} \leqslant \frac{1}{2q_n},$$

since $q_{n+1} \geqslant 2$ for all $n \geqslant 1$. Therefore,

$$\left|\frac{p}{q} - u\right| \geqslant \frac{1}{2q_n} = \frac{1}{2q}$$

and so $|q_n u - p_n| < |qu - p|$.

Assume now that $q_{n-1} < q < q_n$ and write

$$\begin{pmatrix} p_n & p_{n-1} \\ q_n & q_{n-1} \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} p \\ q \end{pmatrix},$$

so that $a, b \in \mathbb{Z}$ by (3.8). Clearly $ab \neq 0$ since otherwise $q = q_{n-1}$ or $q = q_n$. Now $q = aq_n + bq_{n-1} < q_n$, so $ab < 0$; by (3.11) we also know that $p_n - q_n u$ and $p_{n-1} - q_{n-1}u$ are of opposite signs. It follows that $a(p_n - q_n u)$ and $b(p_{n-1} - q_{n-1}u)$ are of the same sign, so the fact that

$$p - qu = a(p_n - q_n u) + b(p_{n-1} - q_{n-1}u)$$

implies that

$$|p - qu| > |p_{n-1} - q_{n-1}u| > |p_n - q_n u|$$

as required. □

We end this section with the uniqueness of the continued fraction expansion.

Lemma 3.4. *The map that sends the sequence*

$$(a_0, a_1, \ldots) \in \mathbb{N}_0 \times \mathbb{N}^{\mathbb{N}}$$

to the limit in (3.10) *is injective.*

PROOF. Let $u = (a_0, a_1, \ldots) \in \mathbb{N}_0 \times \mathbb{N}^{\mathbb{N}}$ be given. Then it is clear that

$$u = [a_0; a_1, \ldots]$$

is positive. Applying this to $(a_1, a_2, \ldots)$ and the inductive relation

$$u = a_0 + \frac{1}{[a_1; a_2, \ldots]}$$

we see that

$$u \in (a_0, a_0 + \tfrac{1}{a_1}) \subseteq (a_0, a_0 + 1).$$

It follows that u uniquely determines a_0. Using the inductive relation again, we have

$$[a_1; a_2, \ldots] = \frac{1}{u - a_0},$$

which by the argument above shows that u uniquely determines a_1. Iterating the procedure shows that all the terms in the continued fraction can be reconstructed from u. □

The argument used in the proof of Lemma 3.4 also suggests a way to find the continued fraction expansion of a given irrational number $u \in \mathbb{R} \smallsetminus \mathbb{Q}$. This will be pursued further in the next section.

Exercises for Sect. 3.1

Exercise 3.1.1. Show that any positive rational number has exactly two continued fraction expansions, both of which are finite.

Exercise 3.1.2. Show that a continued fraction in which some of the digits are allowed to be zero (but that is not allowed to end with infinitely many zeros) can always be rewritten with digits in $\mathbb{N}$.

Exercise 3.1.3. [Dirichlet principle] For a given $u \in \mathbb{R}$ and $n \geqslant 1$ consider the points $0, u, 2u, \ldots, nu$ (mod 1) as elements of the circle $\mathbb{T}$. Show that